INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	

CLASS		INDEX NUMBER	

CHEMISTRY 9729/04

Paper 4 Practical **14 Aug 2018**

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.
Give details of the practical shift and laboratory when appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 18 and 19.
An **insert** is printed on **page 21**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/ 7
2	/ 21
3	/ 27
Total	/ 55

This document consists of **20** printed pages and **2** blank pages.

1 You will find the relative atomic mass, A_r, of magnesium by measuring the volume of hydrogen produced when a known mass of metal reacts with an excess of acid.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

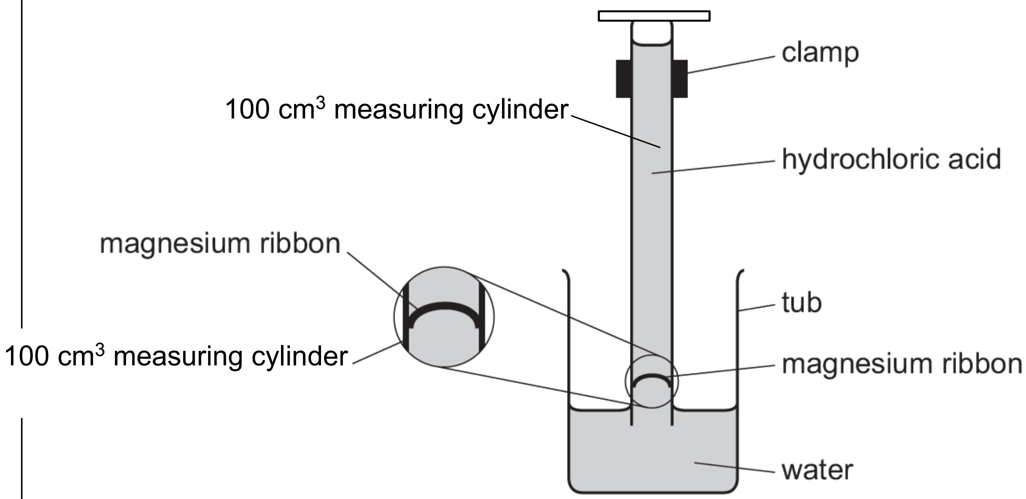
FA 1 is 2.00 mol dm^{-3} hydrochloric acid, HC*l*.

FA 2 is magnesium, Mg.

(a) **Method**

Read through the whole method before starting any practical work.

- Fill the tub with water to a depth of about 5 cm.
- Weigh the magnesium, **FA 2** and note its mass below.
- Fill the 100 cm^3 measuring cylinder to about 30 cm^3 mark with hydrochloric acid, **FA 1**.
- Gently, with minimal disturbance to the surface of the solution, top up the measuring cylinder containing **FA 1** with distilled water to about 4 to 5 cm above the 100-cm^3 mark.
- Bend the **FA 2** strip into a U-shape.
- Carefully place the **FA 2** strip about half a centimeter into the measuring cylinder so that it is above the liquid and friction holds it in position.
- Hold a piece of paper towel over the open end of the measuring cylinder, invert the measuring cylinder and immediately place it in the tub of water. Remove the paper towel and clamp the measuring cylinder as shown in the diagram.



- The liquid level should now be on the scale of the measuring cylinder. If it is not, repeat the above procedure to set up your experiment again.
- Record the initial reading on the measuring cylinder. Remember that the scale is now upside down.
- Leave the apparatus so that the acid from the measuring cylinder diffuses around the **FA 2** and reacts.
- You should start **Question 2 or Question 3** while waiting for the reaction to complete.
- When all the **FA 2** has reacted, note and record the final reading on the measuring cylinder.
- Calculate the volume of hydrogen produced.

Results

Mass of Mg used /g	0.069
Initial reading on measuring cylinder / cm^3	10.0
Final reading on measuring cylinder / cm^3	85.0
Volume of H_2 produced /cm^3	75.0

M1 mass of magnesium, initial and final readings and **volume of hydrogen** with **unambiguous headings** and **correct unit.**
M2 mass recorded to 3d.p. and readings recorded to nearest 0.5 cm^3.

[2]

M1	
M2	

(b) Calculations

Show your working and appropriate significant figures in the final answer to **each** step of your calculations.

(i) Calculate the number of moles of hydrogen produced.
(Assume that 1 mole of gas occupies 24.0 dm^3 under these conditions.)

(If you are unable to calculate the volume of hydrogen produced in **(a)**, you may assume that the volume of hydrogen produced is 70 cm^3 for the calculation here. Note: this is a hypothetic value.)

M3 Correct calculation moles H_2 = volume collected / 24000 mol

Amount of H_2 = 75/24000
= 3.125×10^{-3}
= 3.13×10^{-3} mol

Amount of H_2 = ………………………… [1]

M3	

(ii) Use your answer to **(i)** and the mass of magnesium used to calculate the A_r of magnesium.

M4 Correctly calculation using $A_r = \dfrac{\text{mass used}}{\text{amount of } H_2}$

Ar = $0.0690/ 3.125 \times 10^{-3}$
= 22.08
= 22.1

A_r of Mg = ………………………… [1]

M4	

(c) What would be the effect on the value of the A_r of magnesium calculated if the temperature of the room was much lower than that for your experiment? Explain your answer.

……………………………………………………………………………………………………

		.... [2]	M5 M6	
		M5 Volume (gas) measured/**or** moles/amount gas/H_2 would have been less M6 *correct link betw Ar and amt/vol i.e* A_r greater (conditional: must follow from smaller **moles** of H_2/Mg) **or** M5 Use correct molar volume for new room temperature M6 A_r unchanged (but must follow from V_m smaller)		
	(d)	A similar experiment was repeated using aluminium instead. The reaction was much slower. Explain why aluminium took a longer time to react completely.		
		.... [1]	M7	
		M7 Aluminium has a protective **oxide layer** Or Al is **less reactive** than Mg. Hence, slower.		
		[Total: 7]		

2 Sodium thiosulfate reacts with acid to produce a pale yellow precipitate of sulfur.

$$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow S(s) + SO_2(aq) + H_2O(l)$$

You will investigate how the rate of this reaction varies with the concentration of thiosulfate ions.
To do this, you will measure the time taken for a fixed amount of sulfur to be formed.

FA 3 is 0.10 moldm^{-3} sodium thiosulfate, $Na_2S_2O_3$.
FA 4 is 1.00 moldm^{-3} hydrochloric acid, HC*l*.

(a) **Experiment 1**

- Use the larger measuring cylinder to transfer 40 cm^3 of **FA 3** into the 100 cm^3 conical flask.
- Use the smaller measuring cylinder to measure 25 cm^3 of **FA 4**.
- Pour the **FA 4** into the **FA 3** in the conical flask and start timing **immediately**.
- Stir the mixture in the conical flask once and place the conical flask on top of the printed insert (page **21**) provided.
- Look down through the solution in the conical flask at the print on the insert.
- Stop timing as soon as the precipitate of sulfur makes the print on the insert **just** invisible.
- Record the reaction time to the nearest second.
- Empty, rinse and dry the conical flask so it is ready for use in **Experiment 2**.
- Rinse the sink with tap water to wash away the products of the reaction.

Experiment 2

- Use the larger measuring cylinder to transfer 30 cm^3 of **FA 3** into the 100 cm^3 conical flask.
- Use the same measuring cylinder to add 10 cm^3 of distilled water to the conical flask.
- Use the smaller measuring cylinder to add 25 cm^3 of **FA 4** to the mixture in the beaker and start timing **immediately.**
- Stir the mixture in the conical flask once and place the conical flask on top of the printed insert provided.
- Look down through the solution in the conical flask at the print on the insert.
- Stop timing as soon as the precipitate of sulfur makes the print on the insert **just** invisible.
- Record the reaction time to the nearest second.
- Empty, rinse and dry the conical flask so it is ready for use in **Experiment 3**.
- Rinse the sink with tap water to wash away the products of the reaction.

Experiment 3

Repeat **Experiment 2** using 20 cm^3 of **FA 3**, 20 cm^3 of distilled water and 25 cm^3 of **FA 4**.

Experiments 4 and 5

Choose suitable volumes that will enable you to investigate further the effect of changing the concentration of thiosulfate ions on the rate of the reaction. You should not use a volume of less than 10 cm^3 of **FA 3**.

Results

The relative rate of the reaction can be calculated as shown.

$$\text{rate} = \frac{1000}{\text{reaction time}}$$

In an appropriate format in the space provided, record:

- all measurements of volumes used,
- the reaction time, to the nearest second,
- the relative rate of the reaction, to 3 significant figures, for each of the five experiments

Expt No	Volume of **FA 3** / cm^3	Volume of water / cm^3	Reaction time/ s	Rate of reaction/ s^{-1}
1	40.0	0.0	27	37.0
2	30.0	10.0	36	27.8
3	20.0	20.0	60	16.7
4	25.0	15.0	43	23.3
5	10.0	30.0	136	7.35

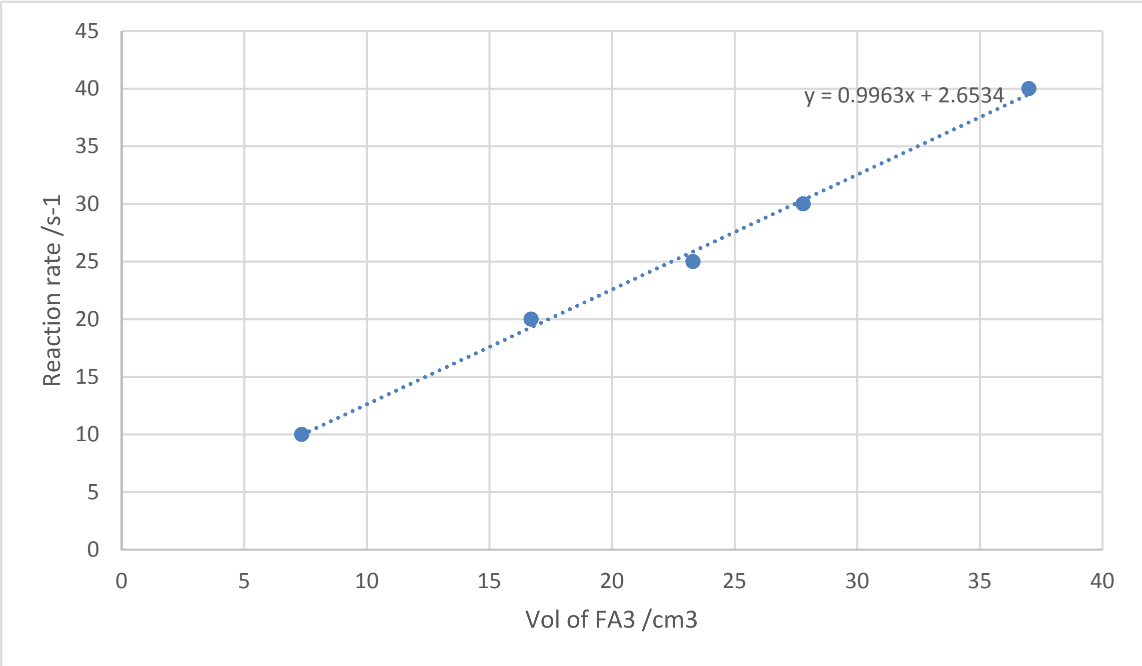
M8– Table that shows volume of FA 3, volume of water, time and rate for 5 experiments. Correct unit for all data: volume(cm^3), time(s), rate(s^{-1})
M9 – all times recorded to the nearest second **(whole number)**, all volume to 1 dp
M10 – Two additional experiments with volume FA 3 not less than 10 cm^3, not more than 40 cm^3 and no volume $\leq$ 2cm^3 close to another volume. Volume of water chosen so that FA 3 + water = 40 cm^3 for the additional experiments. 40cm3, 30cm3, 20cm3 must be present in the table.
M11 – Correctly calculates rate for all experiments and shown to 3 sf.
M12 – Award if candidate's (time for FA3 = 20)/(time for FA3 = 40) is between 1.90 and 2.40.

M8	
M9	
M10	
M11	
M12	

[5]

(b) On the grid, plot a graph of rate of reaction (y-axis) against volume of **FA 3** (x-axis).
Circle any points that you consider anomalous and draw a line of best fit to show how the rate of the reaction depends on the volume of **FA 3**. [3]

M13	
M14	
M15	



M13 – scales that cover more than half the space in both directions and axes labelled correctly with units.
M14 – points plotted correctly. Points must be within half a small square of the correct position.
M15 – line of best fit drawn which ignores anomalous results identified by the candidate

(c) **Use your graph** to calculate the time that the reaction would have taken if 8 cm^3 of **FA 3** had been used. Show **on the grid** how you obtained your answer.

M16 – correct line drawn within 1 small square (horizontal line must be shown) and correctly calculates (1000/ rate)

When 8 cm3 of FA3 is used,
Reaction rate = 10.6

Reaction time = 1000/10.6
= 94.3s

time = ... [1]

M16	

(d) It has been claimed that the reaction is non-zero order with respect to $[S_2O_3^{2-}]$.

State whether you agree or disagree with this claim. Use evidence from your graph to support your answer.

....

....

....

....

.... [2]

M17	
M18	

Agree or disagree (conditional) **M17**

discuss shape of graph/ gradient compared to zero order or finding $t_{½}$ to prove first order and thus is non-zero. **M18**

(e) A student broke the 100 cm^3 conical flask when carrying out the experiment and decided to use a petri dish instead. This has a different shape.

Conical Flask Petri Dish

State and explain what effect this would have on the student's result.

....

....

.... [2]

M19	
M20	

The time recorded will be **longer. M19** **(Conditional; provided understanding that more sulfur is required is shown in explanation)**

More sulfur is required to be formed in order to cover the mark. **M20**

	(f)	A student suggested that the temperature at which the experiment was carried out would also affect the rate of the reaction. Plan an investigation, based on the experiment described in **2(a)**, to determine the effect of temperature on the rate of reaction. You may assume that you are provided with • 0.10 moldm^{-3} sodium thiosulfate, $Na_2S_2O_3$. • 1.00 moldm^{-3} hydrochloric acid, HCl • the equipment normally found in a school or college laboratory Give a step-by step description of how you would carry out the experiment by considering • what you would keep constant in all the experiments, • an suitable number of experiments you would do, and a reasonable temperature range(minimum and maximum temperatures), • the apparatus that you would use in addition to that specified in **2(a),** • the procedure that you would follow and the measurements that you would take.
		....

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		....
		....
		....
		....
		....
		....
		.... [4]
		1. Use the larger measuring cylinder to transfer 40 cm^3 of **FA 3** into the 100 cm^3 beaker. 2. Use the smaller measuring cylinder to measure 25 cm^3 of **FA 4** into another 100 cm^3 beaker. 3. Place both of them in a water bath set at 30°C/ Heat and maintain both solution at 30°C. 4. Ensure that both solutions have reached 30°C by using a thermometer to measure their initial temperature. 5. Pour the **FA 4** into the beaker containing **FA 3** and start timing immediately. 6. Stir the mixture in the beaker once and place the beaker on top of the printed insert provided. 7. Look down through the solution in the beaker at the print on the insert. 8. Stop timing as soon as the precipitate of sulphur makes the print on the insert just invisible. 9. Record the reaction time to the nearest second. 10. Repeat the experiment with the same volume of **FA 3** and **FA 4** used but at temperature 35°C, 40°C, 45°C, 50°C and 55°C.
		M21– at least 5 experiments (not repeats of the same expt) **M22** – experiments covering at least a 25°C range (no greater than 100°C) **M23** – Maintaing the volumes of both reagents to be the same for all experiment. (Marks can be awarded if students repeat same experiment but volume is kept the same) ; **M24**– Indication of **changing** and **maintaining** temperature either via direct heating or water bath.
	(g)	State a hazard that must be considered when planning the experiment and describe precautions that should be taken to keep risks to a minimum.
		.... [1]

		Reference to "hot" apparatus/ sulfur dioxide evolved/ HCl acid (ignore any reference to possible effects) **with** use of **heat proof(thick)** gloves/ use of tongs/ use of fume cupboard. **M25** Spillages and goggles not accepted.
		Shows working in **all** calculations in **1(b)(i) and (ii)**, **2(c).** All calculations must be relevant although they may not be complete or correct. *Any calculation not attempted loses this mark.* **M26** Shows appropriate significant figures (3 ~~or 4 sf~~) in **all** final answers in **1(b)(i)**, **2(c)** and 1 dp in **1(b)(ii)** *Any calculation not attempted loses this mark.* **M27** Shows correct and appropriate units in **all** answers in **1(b)(i) (mol)**, **2(c)(s).** *Any calculation not attempted loses this mark.* **M28**
		[Total: 21]

M21	
M22	
M23	
M24	

M25	

M26	
M27	
M28	

3	**Qualitative Analysis**
	At each stage of any test, you are to record details of the following. • colour changes seen • the formation of any precipitate • the solubility of such precipitates in an excess of the reagent added. Where gases are released, they should be identified by a test, **described in the appropriate place in your observations.** You should indicate clearly at what stage in a test a change occurs. Marks are **not** given for chemical equations. **No additional tests for ions present should be attempted.** **If any solution is warmed, a boiling tube MUST be used.** Rinse and reuse test-tubes and boiling tubes where possible. **Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.**

Question 3 consists of two tasks.

In the first task, you are to explore the chemistry of some compounds of an unknown transition element **W** and determine its identities.

FA 5 is a solid sample of a common dioxide of the unknown transition element **W**,
FA 6 is dilute sulfuric acid, H_2SO_4,
FA 7 is a solid sample of sodium ethanedioate, $Na_2C_2O_4$,
FA 8 is a solution of pure compound **X**, which is the product formed in **(a)(i)**.

(a)	(i)	Test	Observations
		Transfer all of the solid sample of **FA 7** into a small conical flask. Add 25 cm^3 of **FA 6** to this flask.	
		Gently heat the flask until the temperature of the mixture reaches about 60 °C.	White solid $Na_2C_2O_4$ dissolves (completely). [/]
		Swirl the mixture carefully. Place the flask on the heat proof mat.	
		Using a spatula, add **FA 5** to the mixture in small portions. Between each addition, stir the mixture carefully with the thermometer and observe any changes in the temperature of the mixture.	Effervescence/ bubbles after each addition. [/] $C_2O_4^{2-} \rightarrow 2CO_2 + 2e$ Gas produced white ppt in $Ca(OH)_2$ [/]
		Stop adding **FA 5** to the mixture when you think the reaction is complete.	Temperature of mixture rises (even though no heat is applied) [/] Temperature rise stops/ Temperature falls / mixture starts to cool/

			Filter the mixture into a boiling tube and leave the filtrate to stand. The filtrate is compound **X**. Retain this filtrate for use in **(a)(ii).**	effervescence much less vigorous on addition of more **FA 5**. (reaction is complete)[/] 4 - 5[/] – 3 mark 2 - 3[/] – 2 mark 1[/] – 1 mark **M29, M30, M31**
			While you are waiting for (a)(i) to be complete, continue with (c). The colour of the filtrate may change on standing.	
			[3]	
	(a)	**(ii)**	In this part, you are to investigate the effect of the addition of aqueous sodium hydroxide, to separate portions of the filtrate from **(a)(i)** and **FA 8**. In the space below, prepare a suitable table and in it record details of the tests performed and the observations made.	

Test	Filtrate	**FA 8**
To a 1 cm depth of filtrate from **(a)(i)**in a test tube, add **NaOH(aq) (dropwise),** with shaking, until in **excess**.	A light brown ppt is formed and the ppt is insoluble in excess NaOH(aq) [/]. On standing, the ppt turns brown. [/]	A off-white ppt is formed and the ppt is insoluble in excess NaOH(aq) [/]. On standing, the ppt turns brown. [/]
	$Mn^{2+} + 2OH^- \rightarrow Mn(OH)_2$ $Mn(OH)_2 \rightarrow Mn(OH)_3$	

[3]

Both procedure 1 mark (Students can combine both procedure into one) The procedure should describe the addition of a **small volume**/ (dropwise addition), and the addition of an **excess**, of NaOH (aq), both with shaking, to separate samples. **M32**	2[/] – **M33** 4[/] – **M33, M34**

(b) **(i)** Give **two** conclusions you can draw about the nature of the reaction occurring in **(a)(i).**

In each case, support your conclusion by referring to your observations.

Conclusion **1**……………………………………………………………………..

Evidence…………………………………………………………………………..

……………………………………………………………………………………..

M29	
M30	
M31	

M32	
M33	
M34	

……………………………………………………………………………………

Conclusion **2**……………………………………………………………………..

Evidence………………………………………………………………………..

……………………………………………………………………………………

…………………………………………………………………………………… [2]

M35	
M36	

Conclusion 1: the reaction is exothermic.
Evidence: the temperature increased, (without further heating), as **FA 5** was added. **M35**

Conclusion 2: It is a redox reaction
Evidence: Positive limewater test shows that CO_2 is evolved because $C_2O_4^{2-}$ is oxidised (by **FA 5**) **M36**

(ii) Explain why you decided to stop the addition of **FA 5** at the point you chose in **(a)(i)**.

You should support your answer by referring to your observations.

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……………………………………………………………………………………

…………………………………………………………………………………… [1]

Either a convincing explanation based on the cessation of effervescence/bubbles (on addition of more **FA 5)**

OR

A convincing explanation based on the temperature stops rising/starts to fall/remain constant (on addition of more **FA 5)** **M37**

(iii) Consider your observations in **(a)(ii).**

Identify the transition metal ion formed in **(a)(i)** and suggest an explanation for the differences between your observations when you added aqueous sodium hydroxide to the filtrate from **(a)(i)**, and to **FA 8**.

Ion present is …………………………………..

Explanation for difference ………………………………………………………

			.... [1]	M37
			Ion present is Mn^{2+} Some Mn^{2+} in the filtrate is partially oxidised on standing/ in air (to Mn^{3+}, hence resulting in formation of $Mn(OH)_3$.) M38	
		(iv)	In **(a)(i)**, the reaction between **FA 5** and **FA 7** occurs under acidic conditions. Write an ionic equation for this reaction. Use the letter '**W**' to represent the transition metal in this equation.	
			.... [1]	M38
			$\mathbf{W}O_2(s) + 4H^+(aq) + C_2O_4^{2-}(aq) \rightarrow \mathbf{W}^{2+}(aq) + 2H_2O(l) + 2CO_2(g)$ Accept: $2\mathbf{W}O_2(s) + 8H^+(aq) + C_2O_4^{2-}(aq) \rightarrow 2\mathbf{W}^{3+}(aq) + 4H_2O(l) + 2CO_2(g)$ M39	M39

In the second task, you are to perform a series of test to deduce the structure of the unknown hydrocarbon, **FA 9**.

FA 9 is a simple hydrocarbon compound,
FA 10 is bromine water.

(c)	**(i)**	Test	observations
		Place 5 cm^3 of aqueous sodium hydroxide in a test-tube. Add **1 drop** of **FA 9** to this test-tube. Add aqueous potassium manganate(VII), dropwise with shaking, until no further change is seen. Do not exceed 15 drops.	(Purple KMnO4 decolourises) Solution turns green/ blue green [/] Colour deepens as more drops added. [/]
		Note: Eventually, this reaction will produce a brown MnO_2 precipitate. There is no need for you to wait for this to happen.	
	(ii)	Place 5 cm^3 of dilute sulfuric acid in a test–tube. Add **1 drop** of **FA 9** to this test–tube. Add aqueous potassium manganate(VII), dropwise with shaking, until no further change is seen. Do not exceed 15 drops.	Purple **FA 6** decolourise. [/] $MnO_4^- + 8H^+ + 5e \rightarrow Mn^{2+} + 4H_2O$ Colourless solution turns progressively darker brown. [/]
	(iii)	Place 5 cm^3 of water in a test–tube. Add **1 drop** of **FA 9** to this test–tube. Add **FA 10,** dropwise with shaking, until no further change is seen.	Brown bromine water decolourise. (Brown solution turns yellow) [/] 5[/] – 3 marks 3 -4 [/] – 2 marks 1 -2 [/] – 1 mark **M40, M41, M42**
			[3]

Conclusions

(d) Consider the identities of the Mn-containing reactant and eventual product in **(c)(i).**

Suggest a value for the oxidation number of Mn in the coloured ion produced in **(c)(i).** Explain your reasoning.

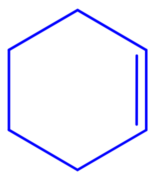
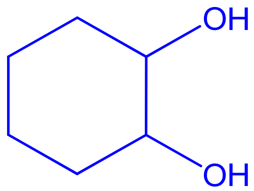
Oxidation number of Mn……………………………..

Explanation……………………………………………………………………………

		……………………………………………………………………………………………… …………………………………………………………………………………………….. [1]	M43
		Allow oxidation numbers of +5 or +6 And An explanation along the lines of - The oxidation numbers of Mn in MnO_4^- and in MnO_2 are +7 and +4 respectively. - So, as the reaction in **(c)(i)** involves a change of oxidation number from +7 to +4, The oxidation number of the coloured ion will be between these value. **M43**	

	(e)	Compound **Y** is the main organic product in **(c)(i)**, when **FA 9** reacts with $KMnO_4$ under alkaline conditions. The molecular formula of **Y** is $C_6H_{12}O_2$.	
		(i)	Deduce the molecular formula of **FA 9**. Explain your deduction. Your explanation must be supported by evidence from your observations in **(c)**.
			Molecular formula of **FA 9** is …………………………………………………… Explanation…………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… [2]
			Explanations based on the following points Molecular formula: C_6H_{10} **M44** • Br_2(aq) decolourised, so C = C present • Cold alkaline MnO_4^- oxidises C=C to C(OH) – C(OH)/1,2-diol • Only two O atoms in molecular formula of **Y**, so only one C =C bond. • So **FA 9** is $C_6H_{12}O_2$ – 2 x OH = C_6H_{10} **M45** any 2 points mentioned
		(ii)	Draw the structural formulae for **FA 9** and compound **Y**. Explain how you deduced the structure of **FA 9**.

M44	
M45	

Structure of **FA 9**	Structure of **Y**
	

				M46 for both structure correctly drawn
			Explanation…………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… …………………………………………………………………………………… [2]	
			Explanation along the lines of • C_6H_{10} has two double bond equivalents • There is only one C=C bond in the molecule So the compound must be cyclic. **M47(idea along this line)**	

M46	
M47	

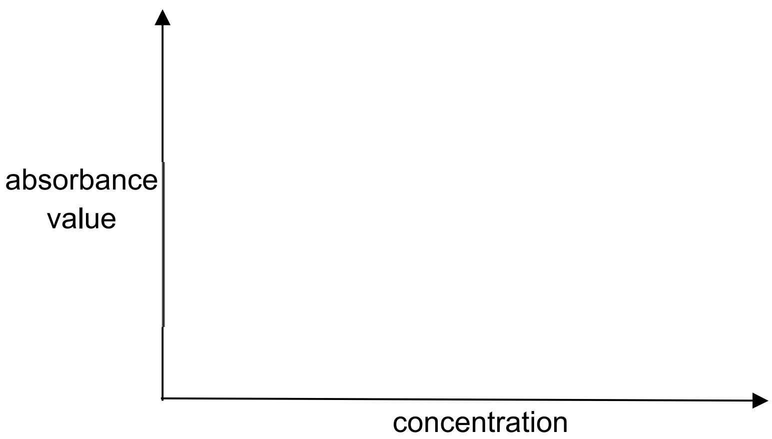
(f) Planning

Many transition metal complex ions are coloured. It is possible to use this property to determine the concentration of a solution of a coloured complex ion. A few cm^3 of the solution is placed inside a machine, known as a spectrometer. This machine measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent. The amount of light absorbed is expressed as an *absorbance value.* The more concentrated the solution, the higher the absorbance value, i.e., the absorbance value is directly proportional to the concentration of the solution.

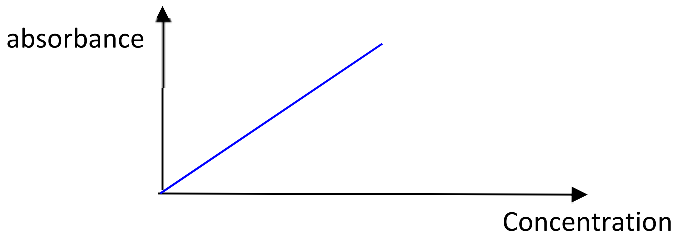
This technique can be used to determine the concentration of a solution of aqueous $\mathbf{WO_4^-}$. A series of known, but different, concentration of solution containing $\mathbf{WO_4^-}$ is prepared. A spectrometer is used to measure the absorbance of each solution. A graph of absorbance against concentration is then plotted. This graph is known as a calibration line.

The experiment is then repeated using a solution of unknown concentration. By comparing the absorbance of this solution with the calibration line, the concentration of $\mathbf{WO_4^-}$ in the unknown solution can be determined.

(i) Sketch the calibration line for the graph of absorbance against concentration that you would expect to obtain.



[1] **M48**



Correct graph **M48**

(ii) Using the information given above, you are required to write a plan to determine the concentration of **W**O_4^- in a solution of **Z**.

You may assume that you are provided with:

- solution **Z**, of unknown **W**O_4^- concentration less than 2 mol dm^{-3};
- a standard solution containing 5.00 mol dm^{-3} of **W**O_4^-;
- access to a spectrometer and instructions for its use;
- graph paper;
- the apparatus and chemicals normally found in a school or college laboratory.

Your plan should include details of:

- the preparation of 100.0 cm^3 of 2.00 mol dm^{-3} aqueous **W**O_4^-;
- the preparation of a suitable range of diluted solutions of accurate concentrations;
- an outline of how the results would be obtained;
- how the calibration line would be used to determine the concentration of **W**O_4^- in solution **Z**.

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[7]

M49	
M50	
M51	
M52	
M53	
M54	
M55	

Preparation of 100.0 cm^3 of 2.00 mol dm^{-3} of aqueous **WO_4^-**

Amount of **WO_4^-** in 100cm^3 of 2.00 mol dm^{-3} **WO_4^-** solution
= (100/1000) x 2.00
= 0.200 mol

Volume of 5.00 mol dm^{-3} standard solution required
= 0.2 / 5.00 = 0.04 dm^3 = 40 cm^3

1. Fill the burette with a standard solution containing 5.00 mol dm^{-3} of **WO_4^-**.
2. Run 40.00 cm^3 of the solution into a 100cm^3 volumetric flask.
3. Make up to the mark with deionised water.
4. Stopper, invert and shake the volumetric flask to obtain a homogeneous solution. Label this solution **Q**.

Preparation of a suitable range of diluted solutions

No	Volume of solution Y/ cm^3	Volume of distilled water / cm^3	Concentration of **WO_4^-** prepared/ mol dm^{-3}	Absorban A
1	20	0	2.00	
2	15	5	1.50	
3	10	10	1.00	
4	5	15	0.50	
5	0	20		

Table 2

5. To prepare 1.50 mol dm^{-3} solution, using a burette, measure out 15.00 cm^3 of solution **Q** into a 50 cm^3 beaker.
6. Using another burette, measure out 5.00 cm^3 of deionised water into the same beaker. Stir/ swirl the beaker to obtain a homogeneous solution.
7. Repeat step 5 and 6 with different volumes of solution Y and distilled water to prepare the stated concentrations of dilute solution.

Outline of how results would be obtained

8. Use the spectrometer to measure the absorbance of each of the diluted solutions prepared and record the absorbance in Table 2.
9. Plot a graph of the absorbance against concentration.
10. Draw the calibration line. (Best fit- linear line going through the origin)

Use of calibration line to determine unknown concentration

11. Using the spectrometer, measure and record absorbance of **WO_4^-** from solution **Z**.
12. Using the graph drawn in step 10, read off the corresponding concentration of **WO_4^-** from the calibration line.

Correct volume (40 cm^3) of 5.00 mol dm^{-3} solution used (BOD if student did not show pre-calculation) **M49**

- **40cm^3 if it is 100cm^3 volumetric flask**
- **100cm^3 if it is 250cm^3 volumetric flask(student must mention in the procedure that they will pour out 100cm^3)**

			Correct procedure for preparing 2.00 mol dm^{-3} solution using a volumetric flask (top up to the mark; stopper; invert and shake) **M50** Use of suitable apparatus with stated capacity **M51** - Burette for measuring out 40cm^3 - Volumetric flask - Mention of appropriate apparatus for dilution Correct dilution concept for preparing diluted solutions (ignore use of less accurate apparatus; allow logical, scientifically – sound preparation of concentrations) **M52** - Using solution of known concentration Suitable concentration of diluted solutions (at least 3 more solutions apart from the 2.00 mol dm^{-3} solution prepared, need not prepare 0 mol dm^{-3}; table not required) Students need to calculate out the concentration in their planning. **M53** Measurement of absorbance of various diluted solutions using spectrometer, and plotting of graph of absorbance against concentration. **M54** Measurement of absorbance of solution **Z** and use of the absorbance value to read off the corresponding concentration in the calibration graph (award if students convey the idea in a graphical diagram) **M55**
			[Total: 21]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	**NH_3(aq)**
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white. ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt. rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ion	***reaction***
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^- (aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^- (aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^- (aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^- (aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^- (aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil; NO liberated by dilute acids (colourless NO → (pale) brown NO_2 in air)
sulfate, SO_4^{2-} (aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-} (aq)	SO_2 liberated on warming with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	***test and test result***
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	***colour of element***	***colour in aqueous solution***	***colour in hexane***
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

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